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(1) Research Program/Project/Study Title: Phenotypic Characterization of Panay Native Cattle in the Province of Capiz	
Proponent Name:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor I
Co-Authors (with academic rank):	Dr. Leah a. Ingalla- Associate Professor III
Campus/Satellite College/Address:	Dumarao Satellite College
Telephone/Cell Phone/Fax Number:	09070287765
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:							
		Province:	Capiz						
		Region:	6						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
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(7) Sector/Commodity Livestock	(8) Discipline/Field of Specialization Large Ruminants
(9A) Abstract This study was conducted to gather information for the production management system, socio-demographic and socio- economic profile of the Panay native cattle raisers in the province of Capiz. Qualitative and quantitative phenotypic characteristics of Panay native cattle were gathered. Qualitative variables includes identification and body skin color, hair coat pattern, muzzle, eyelid and hoof color, presence of horn, horn color and shape, hair type, ear shape and orientation, presence of hump, hump size and position, facial profile, tail type and shape, color of testicles, basic temperament, dewlap size, backline profile, rump shape, and navel flap and preputial sheath size. Quantitative variables documented were body length and weight, height at withers, circumference of heart girth, length of neck, length and width of ear, length of horn, face, muzzle, back, tail and hair, muzzle and hock circumference, width of tail head, pin bone and pelvic bone, length of udder, vulva and teats, and width of udder and teats. Four selected towns, within each municipality, selected four barangays and 25 heads of Panay native cattle were evaluated, a total of 100 heads of cattle from each municipality. Descriptive Statistics and techniques for categorical data analysis were used in the interpretation of the results. Majority of cattle raisers in Capiz are small scale farmers and practiced the Pastoralist way of grassland based system. Native cattle were resistant in several diseases, parasitic infections, ability to prosper in difficult climatic, and nutritional condition. Indigenous Cattle were having a glossy, brown with white and black hair coat, and presence of small hump located on cervico-thoracic. Male cattle are higher compare to female native cattle. (9B) Keywords Panay native cattle, Socio- Economic, Socio- Demographic, Qualitative Traits, Quantitative traits	
(10) Introduction The indigenous cattle are a local breed that were widespread in the Philippines before the Spanish domination of the islands that dates back to the first half of the 16th century. Like all of the cattle breeds in the southern part of Asia, this one is also the fruit of cross breeding between descendants of the <i>Bos Indicus</i> (zebu) and <i>Bos taurus</i> breeds. The Philippine indigenous cattle is small in stature, usually has a yellow or light brown coat and has short horns. The males weigh about 350 kg and have a small hump, while the females weigh a maximum of 230 kg and have no hump. Cattle in the Philippines are kept in both backyard and commercial farms. Based on BAS (2001), about 90% of the cattle production in the country is kept in backyard farms while remaining 10% are raised commercially. Cattle are commonly tethered in nearby fields and vacant croplands during the day to feed on naturally occurring vegetables. Cut-and-carry method of feeding is also practiced particularly during rainy days. Usually, these animals are not provided with housing and during sunny days, trees provide shade to the animals. However, in some instance simple sheds are constructed to protect the animals from the sun. At night the animal is brought near the owner's house (FAO, 2003). The breed is incredibly rustic, strong and resistant, and it is known for the ability to prosper in difficult climactic, health and nutritional conditions, which are all frequent in the Mindanao area in particular. Even the grass in the area is not very nutritious and the water is scarce, or at times excessive due to the rain storms; but this breed of cattle does not suffer from these problems. The calves are extremely strong, standing on their own and nursing directly after they are born. This is a docile breed, and as such the breeders do not dehorn them, nor are the males castrated. The animal is used both for the production of dairy products and for its meat. Today they are pasture raised on small family farms that have two or three heads each.	

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Due to the tendency to import more productive cattle breeds – at the beginning of the 20th century about 40 new breeds were imported to the Philippines – as well as that of cross-breeding in order to improve the animals' characteristics and thus earn better on the market, this native breed of cattle is at serious risk of extinction today. (<https://www.fondazioneslowfood.com/en/ark/-of-taste-slow-food/yellow-cattle/>).

In the country, the livestock and poultry Sub sector is a vital segment in the economy and is firmly integrated into the economic structure of farm village life. Cattle are one of the basic livestock with important role in the subsistence of rural families. It is raised as source of cash in times of needs. Moreover, livestock and livestock products are important engines of economic growth, food and nutrition security (Thornton, 2010).

Indigenous cattle breeds constitute an important reservoir of genetic material which developing nations have failed to give adequate recognition (FAO, 2012). Panay native cattle are one of indigenous cattle in Philippines which are located in Panay Island.

To learn more about this breed, phenotypic characterization is very important, hence this study was conducted.

Objectives of the Study

Generally, the study was conducted to evaluate the phenotypic characteristics of Panay native cattle in the province of Capiz.

Specific Objectives

1. Find out the socio-demographic and economic profile of native cattle raisers.
2. Evaluate the production system of native cattle raising in the province of Capiz.
3. Determine the quantitative and qualitative phenotypic characteristics of Panay native cattle in Capiz.

(11) Materials and Methods

Materials

The materials used in the study include the following: experimental animals (Panay native cattle), tape measure, and survey form.

Methods

Research Design. Purposive sampling was used in the study in identifying the respondents. Top two (2) municipalities in first and second districts with high native cattle population were purposively selected. In each municipality, the top four raisers with highest number of Panay native cattle raised were chosen.

Production Management of Panay Native Cattle

The physical, social and economic environment, as well as the resources required for cattle production was documented at every sample site. This includes the socio-demographic and socio-economic profile of the cattle raisers, support services, membership of any farmer organization, and dispersal program availed. Natural environment of native cattle which are important in production system were production type, level of confinement, climatic modifier, type of holding, feed and water availability, the control of diseases, parasites and other threats to health and mating practice were also noted.

Phenotypic Characteristics of Panay native cattle in the province of Capiz

Qualitative and quantitative phenotypic characteristics of native cattle were gathered in the study. Qualitative variables includes identification and body skin color, hair coat pattern, muzzle, eyelid and hoof color, presence of horn, horn color and shape, hair type, ear shape and orientation, presence of hump,

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hump size and position, facial profile, tail type and shape, color of testicles, basic temperament, dewlap size, backline profile, rump shape, and navel flap and preputial sheath size.

Quantitative variables documented were body length and weight, height at withers, circumference of heart girth, length of neck, length and width of ear, length of horn, face, muzzle, back, tail and hair, muzzle and hock circumference, width of tail head, pin bone and pelvic bone, length of udder, vulva and teats, and width of udder and teats.

Physical and linear measurements were taken only from a representative set of adult animals.

Location of the study. This study was conducted in top four municipality with high population of Panay native cattle in the Province of Capiz.

Respondents and/or Subject of the Study

The respondents of the study were the Panay native cattle raisers/ farms owners in the selected Municipality of the Province of Capiz.

Survey Instrument

Interview schedule were used to gather necessary data from the respondents. It was written in English but translated in Hiligaynon during the survey.

Data Gathering Procedure

The primary data was gathered by means of structured interview schedule, which provides information on phenotypic characteristics of Panay native cattle raisers in the Province of Capiz, Questionnaires were used to gather the needed data. Data gathering focused only on the phenotypic characteristics Panay native cattle in the Province of Capiz.

Statistical Tools and Analysis

Descriptive statistical method such as mean frequency, range and percentage were used to analyze the data of the study.

(12) Results and Discussions

Production Management System Panay Native Cattle

Table 1 shows the production management system and socio demographic and socio economic profile of the Panay naïve cattle raisers in the province of Capiz.

Socio-Demographic and Socio-Economic Profile of the Panay Native Cattle Raisers

The result shows that majority of the Panay native cattle raisers age between 22-59 years old, secondary (53.75%) and Tertiary (23.75%) level, farming (66.25%) was majority livelihood of the Cattle raisers with monthly income of 2,000.00 pesos and above. Many respondents raised native Cattle for about 1-10 years (76.75%), fathers were majority accountable of cattle raising.

Member of Farmer Organization

The study shows that majority (80%) were not member of Farmer organization in relation to Cattle production, 97.50% did not avail Cattle recipient from any agency.

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Problems and Diseases Encountered

The study revealed that pasture (22.50%) or grazing area were major problem and scouring (20%) for disease encountered by native cattle raisers.

Panay Native Cattle Production System

Result of the study shows that Panay native cattle raisers in the province of Capiz practiced Pastoralist (96.25%) of grassland based system, crop-livestock for the mixed production system (76.25%), and backyard/scavenger for landless system. Conformed on BAS (2001), about 90% of the cattle production in the country is kept in backyard farms while remaining 10% are raised commercially.

Level of Confinement and Climate Modifiers

Continuously unconfined and confined only at night (43.75%) got the same level of confinement practiced by native cattle raisers. Shelters and shading trees were basic heat protection.

Control of Diseases, Parasites, and other Threats to Health

Result of the study shows that many of the respondents did not practice vaccination (78.75%) and only few subjected to prevent ectoparasite and endoparasite (22.5% and 21.25% respectively), Veterinary medicine assistance and treatment and also with the traditional treatment of their native when diseases and parasites occur. This simply shows that their native cattle were resistant from any different health threats.

Herd Size and Composition

The study shows that herd size of Panay native cattle composed of bull, cow, heifer and calve. Majority of the herd were heifer (29.80%) and cow (35.86%), and most of the respondents raised 1-5 heads (82.50%) of cattle.

Feed and Water Availability Management

Majority of the cattle raisers in Capiz normally practice the non-restriction of drinking water and feeding of their native cattle, applied salt in drinking water (93.75%) and almost half of the raisers (56.25%) had no year to year variation of feed availability.

Reproductive Strategies

Type of reproductive control

Majority of the raisers practice controlled mating (76.25%), applied hand or pen mating for controlled, none of them used Artificial Insemination, and quarterly for uncontrolled mating (23.75%).

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Table 1. Socio-Demographic and Socio-Economic of Panay Native Cattle Raisers in Province of Capiz.

	Description	Percentage
<i>Socio-Demographic and Socio-Economic Profile of the Respondent</i>		
Age	<6 y/o	
	7 – 12 y/o	
	13 – 16 y/o	
	17 – 21 y/o	
	22-59 y/o	80
	60 & above	20
Education Attainment	None	3.75
	Elementary	15
	Secondary	53.75
	Vocational	3.75
	College	23.75
	Post Graduate	0
Occupation	Farming	66.25
	Laborer	12.5
	Employment	7.5
	Service	0.25
	Fishing	1
	Other	12.5
Monthly Income	<1,000	1.25
	1,001-2,000	11.25
	2,001-3,000	18.75
	3,001-4,000	8.75
	4,001-5,000	10
	5,001-6,000	17.5

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	6,001-10,000	15
	10000 above	17.5
How long raising Cattle (Yr.)	1-10 yrs.	76.75
	11- 20 yrs.	11.25
	20 and above	12
Responsible of raising Cattle	Father	78.75
	Mother	8.75
	Son	3.75
	Daughter	0
	Helper	7.5
	Other	1.25
Member of Farmer Org?	Yes	20
	No	80
Recipient of Cattle	Yes	2.5
	No	97.5
Cattle recipient sources	Government	100
	Cooperation	0
	NGO	0
Problems encountered in raising Cattle	None	37.5
	Pasture	22.5
	Ticks	5
	Others	35
Diseases encountered and treatment	None	60
	Scouring	20
	Lameness	3
	Bloated	1
	Others	16

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Table 2. Production management of Panay Native Cattle in the province of Capiz.

	Description	Percentage
<i>Livestock Production system</i>		
Grassland- based system	Ranching	3.75
	Pastoralist	96.25
Mixed system	Crop-livestock	76.25
	Agro pastoralist	21.25
	Agroforestry	0
	Aqua- Livestock	2.5
Landless System	Industrial	3.75
	Backyard/scavenger	96.25
Level confinement	Continuously unconfined	43.75
	Confined only at night	43.75
	Continuously confined	12.5
Climate modifiers	Basic heat protection	78.75
	Basic cold protection	7.5
	Housing not completely climate controlled	13.75
	Housing completely climate controlled	0
	Cooling Facilities	0
Most animals are vaccinated	Never	78.75
	Occasionally	20
	Regularly	1.25
Subjected to preventive ectoparasites	Never	70
	Occasionally	22.5
	Regularly	7.5
Subjected to preventive endoparasite	Never	70

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	Occasionally	21.25
	Regularly	8.75
Provided by Veterinary treatment	Never	83.75
	Occasionally	12.5
	Regularly	3.75
Provided by traditional treatment	Never	67.5
	Occasionally	30
	Regularly	2.5
Herd Size	1-5 yrs. old	82.5
	6- 10 yrs. old	15
	11- 20 yrs. old	1.25
	21 above	1.25
Herd Composition	Bull	6.56
	Cow	35.86
	Heifer	29.80
	Calve	27.78
<i>Feed and water availability management</i>		
Access to drinking water	Normally not restricted	95
	Occasionally not restricted	5
	Frequently restricted	0
Salt content in drinking water	Yes	93.75
	No	6.25
Feed Availability	Not restricted	68.70
	Frequently restricted during period of the year	31.30
	Restricted throughout the year	0
High yr. to yr. variation of feed availability	yes	43.75

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No		56.25
<i>Reproductive Strategies</i>		
Type of reproductive control	Uncontrolled mating	23.75
	Controlled mating	76.25
Methods employed for controlled mating	Hand or pen mating	100
	Artificial Insemination	0
Phenotypic Characteristics of Panay Native Cattle		
Qualitative Traits		
Many of female cattle has an age ranging from 1-3 years old (36.86%) and 1 month – 1 year old for male (38%), brown with white hair coat color (10.53%) for female and half brown with black for male (50%), no pigment body skin color for female (72%) and pigmented with black for male (60%), patchy body hair coat color pattern for female and male (71.14% and 86% respectively), and mostly were both glossy hair type. Majority of female and male cattle were pigmented muzzle, eyelid, and hoof color. More than half of female and male were having horn presence (63.14% and 66% respectively), straight (33.14% and 64%), and black (52.57% and 58% respectively) in color. Many were having straight edge ear shape, and drooping ear orientation. Almost native cattle having presence of hump located on cervic-thoracic and small size, mostly straight facial profile, fat rump tail type where straight in shape. Half of male were brown and black testicles color. These conform to the study of Piller 2020, that they are vary in color from the usual yellow to white or to brownish orange, but are almost always all one colour. They always have humps, some quite large others smaller. They have small horns or polled. The average weight is 280kg (617lbs) and can be seen in Iloilo or Panay Island. Many of female where docile temperament (84.57%) and 64% for male. More than half of Cattle were small dewlap size, mostly backline profile were slope down from wither, flat rump, half of navel flap of Female where absent, and more than half Male (56%) preputial sheath were medium.		
Table 2. Qualitative Traits of Panay Native Cattle in Province of Capiz.		
QUALITATIVE TRAITS		PERCENTAGE
	Description	FEMALE MALE
Number of Cattle		350 50
Age of Cattle	1m-1 yr	28.86 38
	1.1-3	36.86 32

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	3.1-5	28.00	26
	5.1-8	6.29	4
Hair color	Brown	8.24	16
	Brown with white	10.53	28
	Brown W/ black	8.00	50
	Others (specify)	1.80	6
Skin color	No Pigment	72	40
	Pigmented W/ black	26.86	60
	Other	1.14	
Coat pattern	Patchy	71.14	86
	Plain	12.86	6
	Spotted	16.00	8
Muzzle color	Pigmented	87.14	84
	Not pigmented	13.14	16
Eyelid color	Pigmented	99.71	100
	Not pigmented	0.29	
Hoof colour	Pigmented	88.00	84
	Not pigmented	12.00	16
Horn presence	Present	63.14	66
	Absent	36.86	34
Horn shape	Straight	33.14	64
	Curved	26.86	4
Horn colour	Black	52.57	58
	Brown	7.71	10
	White	0.57	
Hair type	Glossy	86.57	90
	Dull curl	3.14	

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	Curly	1.14	
	Straight	4.00	10
Ear shape	Rounded	5.43	8
	Straight edge	94.57	92
Ear orientation	Erect	2.57	
	Lateral	2.86	6
	Drooping	1.14	2
	Horizontal	93.43	92
Presence of hump	Yes	96.29	98
	No	3.71	2
Hump position	Thoracic	0.86	
	Cervico-thoracic	96.29	98
Hump Size	Small	57.43	52
	Medium	38.57	44
	Large	0.29	2
Facial profile	Straight	87.43	84
	Concave	3.71	8
	Convex	8.86	8
Tail type	Thin	36	38
	Fat rump	57.1	60
Tail shape	Straight	85.43	78
	Cylindrical turn up at end	14.57	22
Colour of testicles	Black		50
	Brown		50
Basic temperament	Docile	84.57	64
	Moderately tractable	17.43	34
	Wild	0.86	2

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Dewlap size	Absent	2.29	2
	Small	53.71	54
	Medium	43.71	44
	Large	0.29	
Backline profile	Straight	17.71	6
	Slope up towards the rump	10.57	14
	Slope down from withers	68.00	78
	Dipped (curved)	3.71	2
Rump	Flat	65.43	72
	Sloping	22	18
	Roofy	12	10
Navel flap (Cow)	Absent	50.86	
	Small	38.29	
	Medium	10.29	
	Large	0.29	
Preputial sheath (Bull)	Absent		38
	Small		56
	Medium		44
	Large		

Quantitative Traits of Adult Panay Native Cattle

The study revealed that the average body weight of male cattle were heavier than female cattle. In terms of body length, circumference of heart girth and length of neck, length of face, muzzle circumference, and length of back the male cattle were bigger and longer. in terms of height at withers, length of tail, width of pin bone, and width of pelvic bone female cattle were bigger and longer than male cattle. In terms of length of ears, width of ears, length of horn, length of muzzle, width of tail head, hock circumference, and length of hair female and male cattle where almost similar in length and size.

The average length and width of female cattles' udder were 13.18 cm and 11.95 cm respectively, 5.54 cm for vulva's length, and 2.95 cm and 1.93 cm for length and width of teats.

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Table 3. Quantitative Traits of Panay Native Cattle in Province of Capiz.

QUANTITATIVE		
	FEMALE	MALE
Body weight	205.11	226.88
Height at withers	112.51	108.31
Body length	110.87	113.31
Circumference of Heart girth	141.79	147.42
Length of neck	33.58	36.14
Length of ear	18.07	18.19
Width of ear	11.66	11.83
Length of horn	12.61	12.11
Length of face	37.77	39.19
Length of muzzle	5.22	5.14
Muzzle Circumference	40.14	42.00
Length of back	94.34	99.67
Length of tail	67.00	65.78
Width of tail head	8.11	8.03
Width of pin bone	19.29	18.67
Width of Pelvic bone	34.97	33.97
Hock Circumference	24.19	24.31
Length of hair	1.33	1.25
Female		
Length of udder	13.18	
Width of udder	11.95	
Length of vulva	6.75	
Length of teats	2.95	

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Width of teats 1.93

(13) Conclusions and Recommendations

Conclusions

The result shows that majority of the Panay native cattle raisers age between 22-59 years old, secondary and Highschool and Tertiary level, farming was majority livelihood of the Cattle raisers with monthly income of 2,000.00 pesos and above. Many respondents raised native Cattle for about 1-10 years fathers were majority accountable of cattle raising.

Panay native cattle raisers in the province of Capiz practiced Pastoralist of grassland based system, crop-livestock for the mixed production system, did not practice vaccination and few were subjected to prevent ectoparasite and endoparasite, practice the non-restriction of drinking water and feeding of their native cattle, applied salt in drinking water, and practice controlled mating by hand or pen mating.

Majority of Panay native cattle were having a glossy hair type, brown with white and black for hair coat color, straight horns, straight edge ear shape, and drooping ear orientation, having presence of hump located on cervic-thoracic and small size, mostly straight facial profile, small dewlap size, mostly backline profile were slope down from wither, flat rump, half of navel flap of female where absent, and more than half male preputial sheath were medium.

Average body weight of male cattle were heavier than female cattle. In terms of body length, circumference of heart girth and length of neck, length of face, muzzle circumference, and length of back the male cattle were bigger and longer. in terms of height at withers, length of tail, width of pin bone, and width of pelvic bone female cattle were bigger and longer than male cattle. In terms of length of ears, width of ears, length of horn, length of muzzle, width of tail head, hock circumference, and length of hair female and male cattle where almost similar in length and size

Recommendations

Concerned agencies should give focus on the training, knowledge dissemination and financial aspects of native cattle raisers by extending them credit assistant and other financial supports which motivate them to work and ensure profit. Technical assistance to native cattle raisers for the proper management including proper sanitation, feeding, pasture development, diseases control and prevention.

Supporting the dispersal program given by some municipalities is needed to improve the management practices of the native cattle raisers.

Promote and create awareness on how to preserve the Panay native cattle breed in the province of Capiz. Implementing innovative way of raising native cattle to the farmers should be implement to improve indigenous cattle production.

Conduct more studies related to Panay native cattle genetics, feeds and conservation to follow up this study.

(14) Output

- Knowledge and information on the phenotypic characteristics of Panay Native cattle found in the province of Capiz.
- Research paper for scientific conferences and journal publication
- IEC Materials for the Phenotypic characteristics of Panay native cattle of Capiz.

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(15) Technology Transfer Activity Conducted

The result of this study will serve as a bench mark information for the policy maker, and researchers on the characteristics of Panay Native Cattle in Province of Capiz.

(16) Beneficiaries

- Panay native cattle raisers
- Department of Agriculture/ BAI/ PNAD
- Local Government Units

(17) Economic Analysis

(17) Acknowledgments

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